

Background

Random Variables

A real-valued *Random Variable* is a measurable function $X : E \rightarrow \mathbb{R}$ from a probability space $(E, \mathcal{A}, \nu)$ to the measure space $(\mathbb{R}, \mathcal{B}, \lambda)$.

Let μ_X be the push-forward $X^*\nu = \nu X^{-1}$. We call μ_X the distribution of X , denoted by $X \sim \mu_X$. This means for all $B \in \mathcal{B}$,

$$\nu(X^{-1}(B)) = \mu_X(B).$$

By the Law of Unconscious Statistician,

$$\mathbb{E}_{X \sim \mu_X}[X] = \int_{\mathbb{R}} x \, d\mu_X(x).$$

More generally, if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a measurable function, then

$$\mathbb{E}_{X \sim \mu_X}[f(X)] = \int_{\mathbb{R}} f(x) \, d\mu_X(x).$$

In case μ_X is absolute continuous with respect to λ , we let

$$p_X = \frac{d\mu_X}{d\lambda} \in L^1(\mathbb{R}, \lambda)$$

be the Radon-Nikodym derivative, and denote by $X \sim p_X$ with

$$\mathbb{E}_{X \sim p_X}[f(X)] = \int_{\mathbb{R}} f(x) p_X(x) \, d\lambda(x)$$

for $f : \mathbb{R} \rightarrow \mathbb{R}$ measurable. p_X is the density function.

If X is a random variable with values in a measure space, that is, $X : E \rightarrow \Omega$ is a measurable function from a probability space $(E, \mathcal{A}, \nu)$ to a measure space $(\Omega, \mathcal{M}, \mu)$. Let $\mu_X = X^* \nu$ be the push-forward probability measure on Ω , which is denoted by $X \sim \mu_X$.

X itself does not have an expected value in general. If $f : \Omega \rightarrow (\mathbb{R}, \mathcal{B}, \lambda)$ is measurable, then

$$\mathbb{E}_{X \sim \mu_X}[f(X)] = \int_{\Omega} f(x) d\mu_X(x).$$

Moreover, if μ_X is absolute continuous with respect to μ , then we let

$$p_X = \frac{d\mu_X}{d\mu} \in L^1(\Omega, \mu)$$

be the Radon-Nikodym derivative, and we have, for $f : \Omega \rightarrow (\mathbb{R}, \mathcal{B}, \lambda)$ measurable,

$$\mathbb{E}_{X \sim p_X}[f(X)] = \int_{\Omega} f(x) p_X(x) d\mu(x).$$

Entropy

Definition. Let $(\Omega, \mathcal{M})$ be a measurable space and P and Q be two probability measures on Ω . Suppose P is absolute continuous with respect to Q (ie. $P \ll Q$). The Kullback-Leibler divergence (or K-L divergence, relative entropy), is defined as

$$\text{KL}(P \parallel Q) = \int_{\Omega} \log \left(\frac{dP}{dQ}(x) \right) dP(x),$$

where $\frac{dP}{dQ} \in L^1(\Omega, Q)$ is the Radon-Nikodym derivative.

More generally, suppose P and Q are both absolute continuous with respect to another measure μ (ie. $P \ll Q$ may not hold), we let $p = \frac{dP}{d\mu}$ and $q = \frac{dQ}{d\mu}$, both of which are in $L^1(\mu)$, and define

$$\text{KL}(P \parallel Q) := \int_{\Omega} \log \left(\frac{p(x)}{q(x)} \right) p(x) d\mu(x).$$

Note that such a measure μ always exists since we can put $\mu = \frac{P+Q}{2}$.

Note that $\text{KL}(P \parallel Q) \geq 0$ because of Gibbs' inequality: Suppose $\{p_1, p_2, \dots, p_n\}$ and $\{q_1, q_2, \dots, q_n\}$ are discrete probability densities. Then

$$\sum_i p_i \log q_i \leq \sum_i p_i \log p_i,$$

with equality if and only if $p_i = q_i$. Hence, $\text{KL}(P \parallel Q) = 0$ if and only if $P = Q$ as measures or $p(x) = q(x)$ μ -a.e.

Definition. Let P and Q be as above. Let $M = \frac{P+Q}{2}$. The Jensen-Shannon divergence is defined as

$$\text{JS}(P \parallel Q) := \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M).$$

Note that $\text{JS}(P \parallel Q) \geq 0$. It is known that $\text{JS}(P \parallel Q) \leq \log 2$.